newest processors, providing the best performance. However, these come at a higher cost and are not necessary for other environments. In contrast, non-production environments can utilise more affordable machine types (like E2 series) that provide reasonable performance at a significantly lower cost.

Spot VMs (virtual machines): For applications that are fault-tolerant, we can also consider the option of spot VMs. This option provides preemptible instances which can be up to 91% cheaper compared to regular instances.

Scheduled downtime: If spot VMs are not suitable, another option is to explore the possibility of having a scheduled shutdown of the VMs during unused or non-business hours (e.g., six hours every night or on weekends). Adopt this plan only if the scheduled downtime is applicable and doesn't impact the teams using this environment.

VM time to live: GCP also provides an option to specify the maximum runtime of a VM, during VM creation. The VM automatically terminates once the specified time limit is reached. This feature is particularly helpful for temporary workload deployments, as the VMs get automatically deleted after a certain time, preventing unnecessary costs due to idle machines.

• Production environments

Production environment deployments are usually the most cost-intensive as each application requires a significant number of instances to support the large traffic volumes. So, any optimisation effected here will have a significant impact on savings.

Right-sizing: When the lift-and-shift approach is followed, the VM configurations (CPU, memory per instance, number of instances, etc) are replicated from on-premises. However, this is mostly sub-optimal and has a good scope for improvement. If we monitor the resource usage post migration, we will find the instances are mostly underutilised. So, the first step is to right-size your VM instance. Run performance tests and identify the right VM configuration and machine type for the required traffic load. Application characteristics play a crucial role in determining optimal machine type and configuration. Applications that serve API requests to end customers may need high CPU power whereas batch processing applications may need higher memory. Select the appropriate machine type and compute configuration based on your application's needs.

Instance count: GCP has machine types that offer the latest and fastest processors (like N4) as well as specialised ones (like compute-optimised) that might suit certain applications. Pick the right machine type, arrive at the optimal VM configuration, and identify the load supported per instance. Then derive the number of instances needed to support the production traffic. With this approach, you will see a reduced config per VM and a need for a smaller number of overall instances.

Autoscaling: One more practice is to migrate with a plan of fixed number of instances. For most enterprise applications, the load is not constant and follows patterns. Peak load is seen in business hours and there's very less load in non-business hours. So, provisioning a fixed number of instances based on peak load will lead to wastage of resources during non-peak hours. To optimise this, we can make use of the autoscaling feature of GCP Managed Instance Groups (MIGs) so that instances can be increased and decreased based on the load. Since costs are based on the compute runtime usage, this will save costs considerably as less instances will be running in non-peak hours.

• Other options for consideration

Discounts: GCP offers another great option of committed use discounts (CUDs). With CUDs, we can obtain great discounts in pricing over multiple cloud resources if we give spend commitments – for example, a commitment to use and pay for a certain amount for one or three years. The discounts are usually higher for a three-year commitment. For example, certain machine types get a 28% discount with a one-year commitment and a deep discount of 46% for a three-year commitment.

Cloud native approaches: In on-premises data centres, the VMs that run batch processing workloads are up and running all the time even though the batch processing jobs run only for a small duration of time. When these are lifted-and-shifted to the cloud and the cloud VMs are run the whole day, it results in a wastage of resources. The application and the VM instance keep running and are charged for even when there are no batch processing jobs being executed. Such workloads need to be redesigned so that the charge is only for the duration of the job. GKE CronJob, serverless mechanisms like Cloud Functions, Cloud Run, etc, can be used to limit costs to only the execution time.

Cloud logging

Applications migrating to GCP integrate with a cloud logging service to make good use of features like log explorer, logs-based metrics, etc, that help with troubleshooting. The costs of cloud logging are mainly based on the volume of data ingested. And if an organisation's applications have not been logging the right volume of data on-premises, the logging costs after migration may come as an unpleasant surprise. Let us look at a few ways to save costs while making use of the features of cloud logging.

Environment-based decision: Cloud logging capabilities are much needed in production environments, but are not as necessary in others, such as development. So, look at the logging costs incurred in non-production environments. If they're high, consider disabling cloud logging integration.